Applications

1

a $f(t) = 250 + 100 \sin 2\pi t$

b $g(t) = 350 - 100 \sin 4\pi t$

c 36 times

d $\frac{1}{8}$ years

e $t = \frac{7}{12}$

2

a $f\,(t) = 150 + 40 \sin 2\pi t$

b $g\,(t) = 100 - 40 \sin 3\pi t$

c 27 times

d $t = 1.70$

3

a Sine, the graph's initial value is at the midline.

b $I = \sin\left(\frac{\pi t}{2}\right)$

c 210 times

4

a 6 seconds

b $2500\ \mathrm{mL}$

c $200\ \mathrm{mL}$

d 60° per second

e $c = \frac{1}{2}$

5

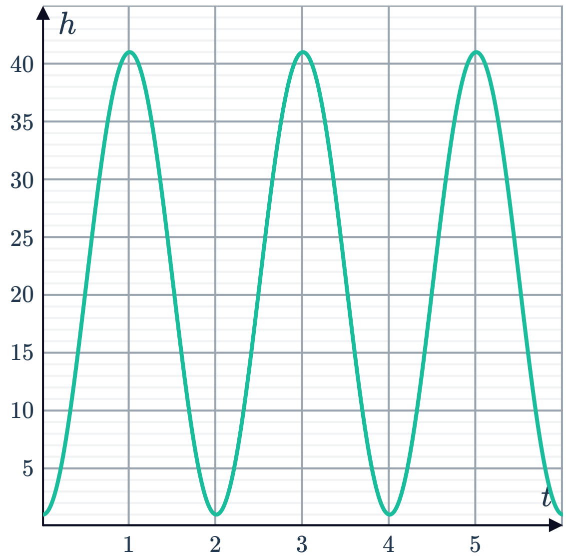
a

i No **ii** No **iii** No **iv** Yes

b

i 2 minutes **ii** 1 m **iii** 41 m

c



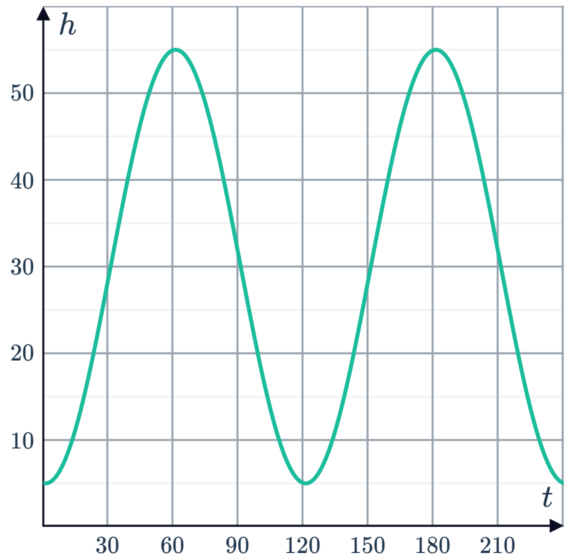
d $h = -20\cos(\pi t) + 21$

e $t = \frac{1}{2}$ minutes

f The height of the passengers above or below the centre of the wheel at time t.

6

a



b 55 m

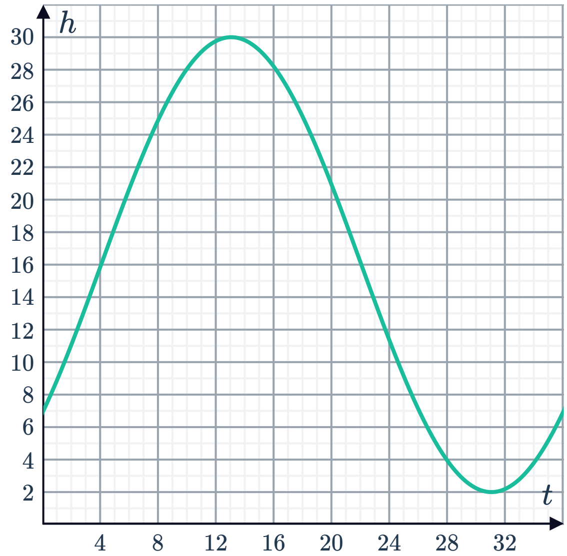
c 5 m

d 36.47 m

e 120 seconds

7

a



b 7.00 m

c $t = 34.82$ seconds

d 3

e 3.88 m

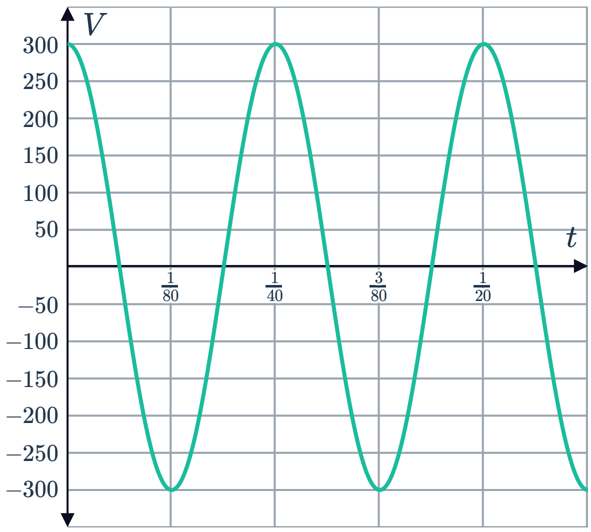
8

a

i Yes **ii** No **iii** No **iv** No

b $V = 300 \cos(80\pi t)$

c



d It would decrease the period of the function and compress the graph horizontally.

9

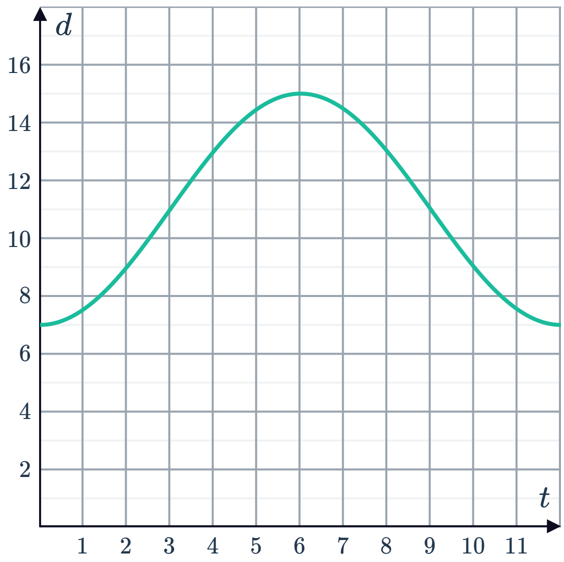
a

Time (t, hours)	0	3	6	9	12
Tide level (d, metres)	7	11	15	11	7

b 4 m

c 12 hours

d



e

i Yes **ii** No **iii** No **iv** No

f $d = -4\cos\frac{\pi}{6}t + 11$

g 2.66 hours

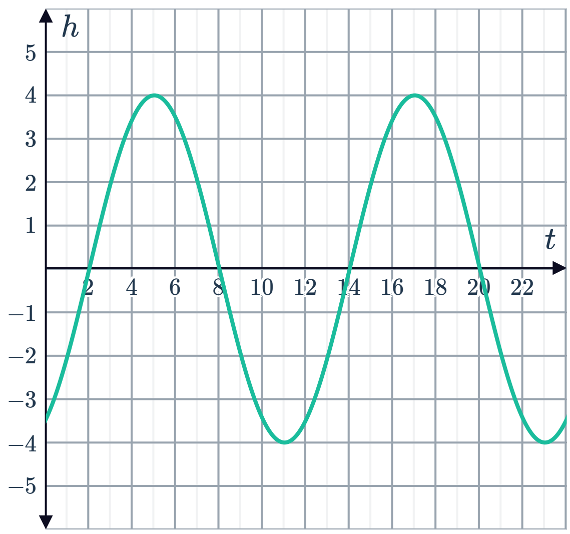
10

a -2 m

b 5:00 am

c 6 hours

d

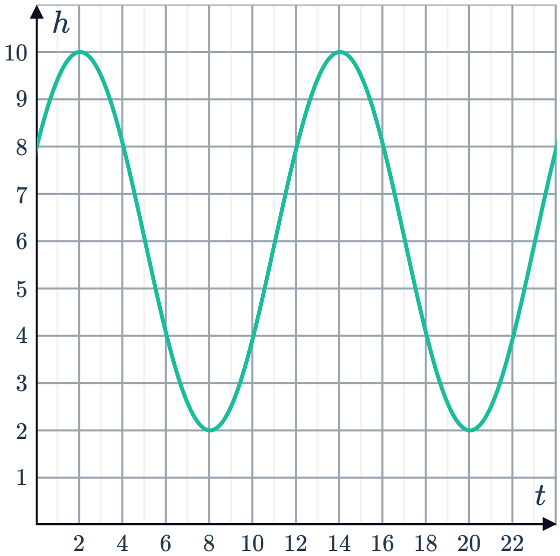


e 0 m

f 0.14 m

a $h(t) = 6 + 4\cos\left(\frac{\pi}{6}(t-2)\right)$

b



c 2:00 am

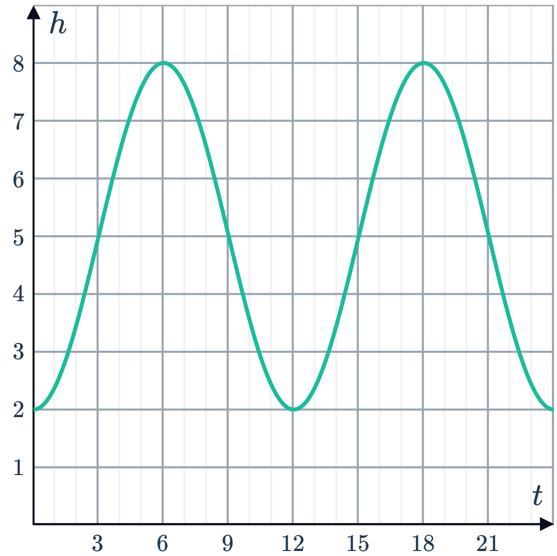
d 8:00 am

12

a

i $d = 5$ m **ii** $a = 3$ m **iii** $b = 30^\circ$/h **iv** $c = 90^\circ$

b



c 6 and 18 hours after noon

13

a

i $a = 40$ cm **ii** $d = 10$ cm **iii** $c = \frac{1}{4}$

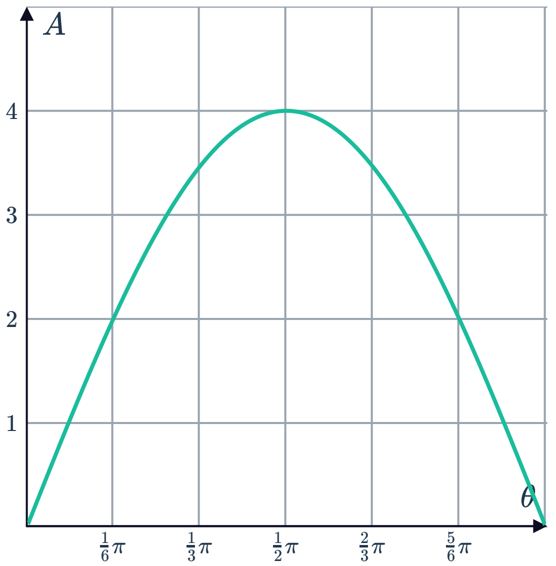
b $t = \frac{1}{3}$ seconds

14

a $A = 4\sin\theta$

b $0 < \theta < \pi$

c



d $\frac{\pi}{6}$

15

a 5 cm to the right of the centre

b To the left

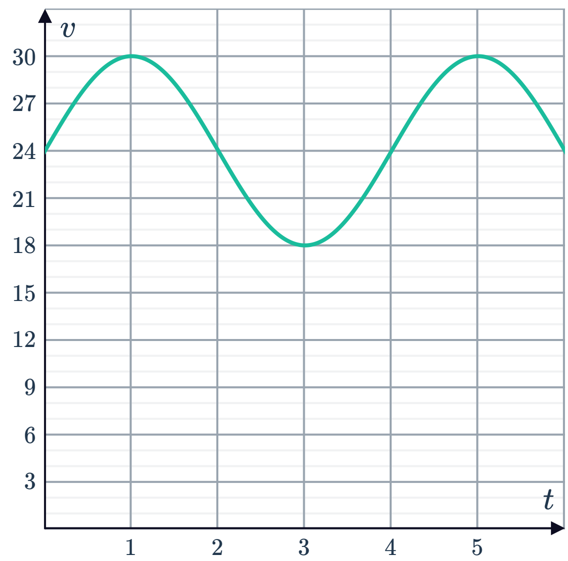
c 1 second

d 4 seconds

e $x = 5\cos\left(\frac{\pi t}{2}\right)$

16

a



b 30 units

c The number of seconds that have passed since the beginning of the section.

d 24 units

17

a 1.44 seconds

b 12.46 hours

c 110 days